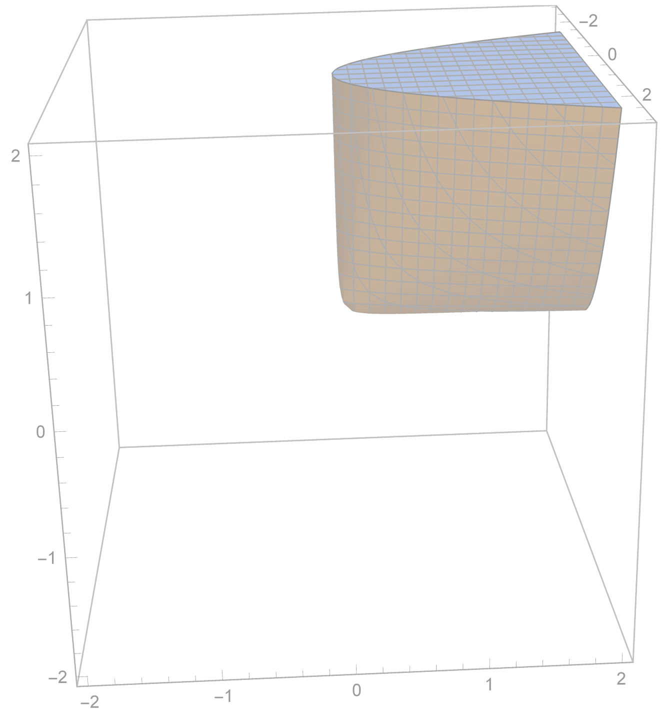
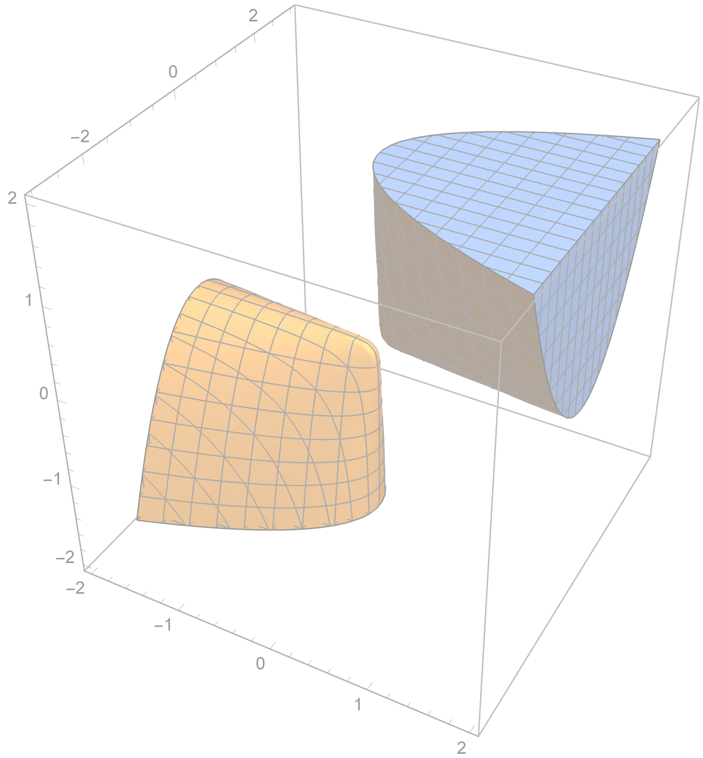
```
RegionPlot3D[x * z - y^2 > 0 && x + z > 0, {x, -2, 2}, {y, -3, 3}, {z, -2, 2}, PlotPoints → 40]
(* this is an SOCP representable set. Notice we need the constraint x+z≥0 *)
```

Out[15]=



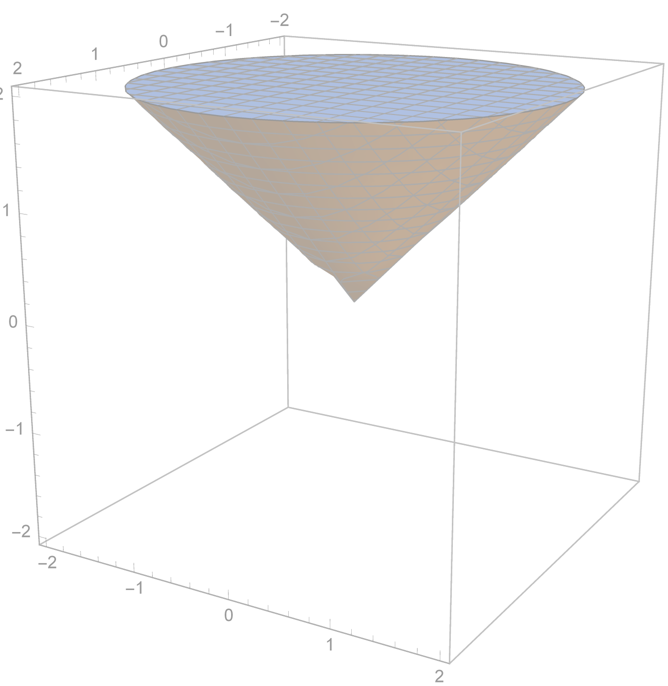
```
RegionPlot3D[x * z - y^2 > 0, {x, -2, 2}, {y, -3, 3}, {z, -2, 2}, PlotPoints → 40]
(* this is not an SOCP representable set,
because it is not a convex set. Note it has two cones,
because we did not enforce x+z≥0. *)
```

Out[18]=




```
RegionPlot3D[x^2 + y^2 - z^2 < 0 && z > 0, {x, -2, 2}, {y, -2, 2}, {z, -2, 2}]
(* This is the basic socp cone. Note again z≥0 *)
```

Out[14]=



```
RegionPlot3D[x^2 + y^2 - z^2 < 0 , {x, -2, 2}, {y, -2, 2}, {z, -2, 2}]
(* this is what happens if we don't have z≥0. Then, the set contains two cones,
one for z≥0, the other for z≤0. therefore, this is a non-convex set, not the SOCP cone.*)
```

Out[16]=